GlobalBuildingAtlas (TUM GBA) and Google 2.5D Temporal Dataset Use Case Documentation

Operational Value Assessment

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1. Executive Summary

Building Footprint datasets are proxies for population estimation to support operational activities. This report assesses the operational value of the TU Munich´s GlobalBuildingAtlas (GBA) and Google’s Open Buildings 2.5D temporal Datasets, for building-level population estimation and humanitarian planning for MSF. Through a comparative analysis, these datasets are analyzed against OpenStreetMap (OSM) across three areas of interest: Tawila, Sudan; Budapest, Hungary; and the Sobat Corridor, South Sudan. The African AOIs (Tawila and Sobat Corridor) were selected to represent operating conditions relevant to MSF's use cases, while Budapest represents a well training-data European benchmark for the GBA model.

Across all three AOIs, one finding held consistently: GBA provides near-complete building height data (100%), followed by Google 2.5D(12 -71%) while OSM provides essentially none (0–1.6%). This represents GBA's clearest and most reliable operational advantage, height value estimation availability. This height value should, however, take into consideration the Root Mean Square Error for GBA, different per region and the Mean Absolute Error for Google 2.5D, 1.5m globally.

Footprint performance, by contrast, varied by AOI. In Budapest, GBA's footprints closely tracked OSM's, consistent with OSM serving as GBA's primary fusion input in well-mapped European regions. In Tawila, OSM's community-mapped footprints appeared more granular than both GBA and Google 2.5D, which showed signs of merging adjacent buildings into single polygons. In the Sobat Corridor, both GBA and Google 2.5D identified substantially more buildings than OSM, suggesting genuine completeness gains in an area with limited existing footprint coverage.

An independent technical check was also carried out to test how GBA handles buildings that span its 5°×5° distribution tile boundaries, since this was not documented in the source paper.

Buildings crossing tile edges were found to be duplicated safely, with identical IDs and geometry in both tiles, resolvable through a straightforward deduplication step.

Taken together, these findings confirm that both datasets' operational value is highest as a height-data source, and as a footprint source specifically in areas where existing OSM coverage is sparse or absent. Known limitations, include GBA´s lack of validated data from Africa while Google 2.5D's published 1.5 MAE has one independent test in Nairobi, Africa, results to 2.5m MEA (Watson Cs, 2025). GBA´s dataset's 2019-era imagery vintage, should inform how these findings are applied to time-sensitive or high-precision operational decisions. In these cases, Google 2.5D presents a temporal advantage of spanning from 2016 -2023. As of date, there are no announced plans for the dataset upgrade. However, the institute claims that the datasets generation pipeline described in the paper and code available on the GitHub page supports rapid and scalable update with available PlanetScope imagery.

2. Introduction

Medecins Sans Frontieres (MSF) workflow currently utilizes open-source building footprint datasets from OSM, Microsoft Buildings as well as Google Open Buildings, to approximate population in areas of operation. Complimented by gridded population data, this provides critical information on population distribution and density helping MSF's operations such as vaccination campaigns, water and sanitation support.

New datasets, with height parameters, present a possibility to enhance the accuracy of population estimation. This is where the TU Munich GBA published in 2025 and Google Open buildings 2.5D temporal datasets comes in.

GlobalBuildingAtlas (GBA) is an open, global dataset, developed by the Technical University of Munich, providing building footprint polygons (GBA.Polygon), 3m-resolution height maps (GBA.Height), and Level of Detail 1 3D building models (GBA.LoD1), derived from PlanetScope satellite imagery. It combines the authors' own machine-learning pipeline with a quality-guided fusion of existing sources (OSM, Google Open Buildings, Microsoft, CLSM).

Unlike TUM GBA, which is a vector, Google Open Building 2.5D temporal dataset is a raster dataset containing data about building presence, building centroids, and building heights at an annual cadence from 2016-2023. It is produced from open-source, low-resolution imagery from the Sentinel-2 collection. The dataset is available across Africa, South Asia, South-East Asia, Latin America, and the Caribbean. (Wojciech Sirko, 2024)

3. Dataset Overview

3.1 GBA.Polygon

The United Nations believes that there are 4B buildings on Earth. Global building footprint polygons claim to cover approximately 2.75 billion buildings. This makes the dataset the most complete global count to date. Below is a table showing the existing large scale building footprint products, their coverage and number of buildings.

Product	*Coverage*	*No. Of Buildings (Billions)*
Microsoft	*Global excl. China*	*1.4*
OSM	*Global*	*0.45*
Google Open Buildings	*Global South*	*1.8*
CLSM	*East Asia*	*0.28*
3D GloBFP	*Global*	*1.7*
TUM GBA.Polygon	*Global*	*2.75*

These Open-Source footprints (OSM, Google Open Buildings, Microsoft Buildings and CLSM), together with GBA's PlantScope derived polygons, fused to form the GBA-polygon.

3.2 GBA.Height

Global building height raster at 3m resolution, roughly 30x finer than prior global products (90–150m).

3.3 GBA.LoD1

The Level of Detail 1(LoD1) datasets are simplified 3D representations that capture the basic shape and height of each building. This building model combines GBA.Polygon footprints with GBA.Height, covering ~2.68 billion buildings with 97.7% height completeness.

3.4. Google 2.5D Temporal Dataset

The Open Buildings 2.5D Temporal Dataset contains annual data spanning eight years (2016-2023) with building presence, fractional building counts, and building heights covering approximately 58 million square kilometers. It was produced using publicly available, low-resolution imagery from the Sentinel-2 satellite mission, which has approximately 5-day revisit times and world-wide coverage. The resulting dataset has an effective spatial resolution of 4 meters, equivalent to what could be achieved by a high-resolution model using a single frame of 4-meter resolution imagery

3.5. OSM

OpenStreetMap is a global, volunteer contributed building footprints with mapping ongoing since 2008. Unlike GBA, OSM buildings are digitized by community mappers. This results in higher positional and shape accuracy in mapped areas. The datasets are also dynamic and updated frequently while GBA is static.

4. Methodology

4.1 AOI Selection

Areas of interest for this analysis were selected based on two regions. Tawila, Sudan, and Malakal, South Sudan, are locations MSF has operational projects, while Budapest, Hungary represents a European city, where the training and test data were collected. The latter was selected to test GBA model performance in a region where it's expected to perform better.

4.2 Tools and Pipeline

Data was accessed and processed using a Python/Jupyter workflow.

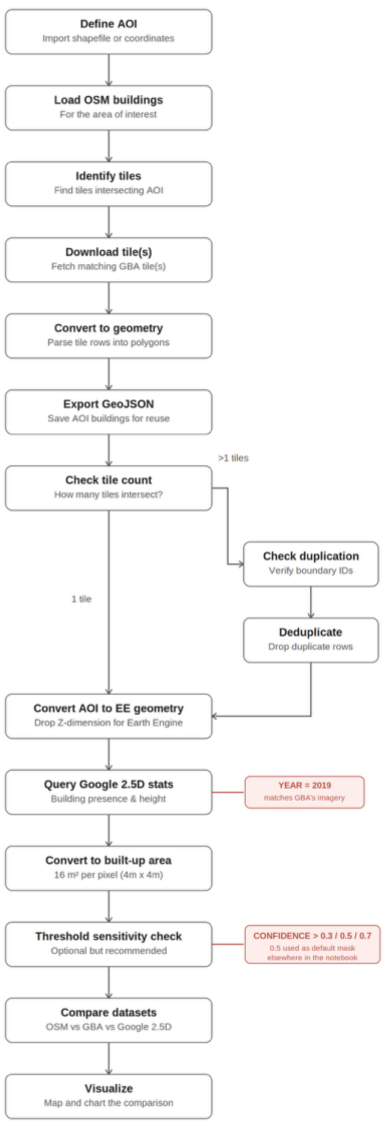

Figure 1: Notebook workflow. Full notebook in Appendix A

This Notebook compares the three building footprints datasets OSM, GBA and Google Open Buildings 2.5D Temporal, queried via Google Earth Engine. After the AOI is defined by either coordinates or importing shapefile, OSM buildings are extracted by quarrying the API. For GBA dataset tile accessed from a mirror repository. The downloaded parquet file is stored and accessed in the notebook locally. The AOI geometry created in the first step is reused to query Google 2.5D, avoiding the need to redefine the area a second time.

To maintain consistency, since GBA timeline is 2019 Planetscope imagery, Google 2.5D also accessed 2019 sentinel 2 imagery.

Two parameters in this section were deliberately chosen rather than left at arbitrary defaults, and are worth calling out explicitly: **the query year is set to 2019**, matching GBA's own PlanetScope imagery vintage as closely as possible, so that any comparison between the two datasets reflects the same approximate point in time rather than comparing, say, 2019 GBA data against 2023 Google 2.5D data. **The building-presence confidence threshold is set to 0.5** when masking pixels to compute a "building-only" mean height and built-up area — this is the standard midpoint for converting Google's continuous 0–1 confidence score into a binary building/non-building classification, but it is an analyst choice, not a value specified by Google. To make this explicit rather than hidden, the notebook includes an optional sensitivity check that recomputes the built-up area using 0.3, 0.5, and 0.7 as alternative thresholds, so the reader can see how much the result depends on this choice before treating any single figure as definitive.

4.3 Technical Validation Performed

The GBA paper and accompanying GitHub documentation describe the dataset's polygon fusion process as occurring at the administrative-boundary level, prior to the final export into 5°×5° distribution tiles. However, neither source explicitly documents how buildings whose footprints straddle a tile boundary are handled in this final export step — a gap with direct implications for any building count, footprint area, or volume total calculated for an AOI spanning multiple tiles.

To resolve this independently, we constructed a test AOI deliberately centered on a tile boundary (lat 47.5°N, lon 15.0°E, Austria), ensuring the area would draw buildings from two adjacent GBA tiles. After downloading and combining both tiles, we identified all building id values appearing in more than one tile and compared their geometries directly.

Result: buildings crossing the tile boundary are duplicated safely. Thus, the same id and identical geometry appear in both tiles, rather than being split into differently clipped fragments or assigned separate IDs. This means GBA's tiling step does not corrupt individual building footprints at tile edges, and any resulting double-counting in aggregate statistics can be fully resolved with a straightforward deduplication step prior to analysis.

This finding is treated as a confirmed, low-risk characteristic of the dataset rather than a limitation, and the deduplication step has been incorporated into our standard processing pipeline.

5. Area of Interest Case Studies

This section presents building-level comparisons between OSM and GBA for each area of interest, evaluating GBA's operational value in contexts where OSM coverage may be incomplete.

5.1 Area of Interest: Tawila, Sudan

5.1.1 Site Context

Tawila locality in North Darfur hosts one of the largest internally displaced populations in Sudan. Displacement numbers over 715,000 as of May 2026 (IOM, 2026). Apart from the humanitarian situation in the area, this AOI was selected based on the presence of well-mapped OSM buildings. This provides a good comparison of use cases.

Coordinates: 13.487, 24.855

5.1.2 Data Comparison

Summary of OSM, GBA and Google 2.5D building counts and footprint areas for this AOI:

Metric	OSM	GBA	Google 2.5D
Building count	153,530	23,763	44,065
Total footprint area (km²)	1.3	0.5	0.7
Buildings with height data	0%	100%	12%
Mean Height(m)	-	0.6	4.1

Note: the OSM building count (153,530) and the correspondingly low GBA mean height (0.6 m) for this AOI are unusually high/low relative to the other AOIs assessed in this report and are flagged here as requiring verification rather than treated as confirmed findings. A likely explanation is that the raw OSM extract for this AOI was not filtered to Polygon/MultiPolygon geometries before counting, which would inflate the count with non-footprint Point features.

5.1.3 Visual Comparison

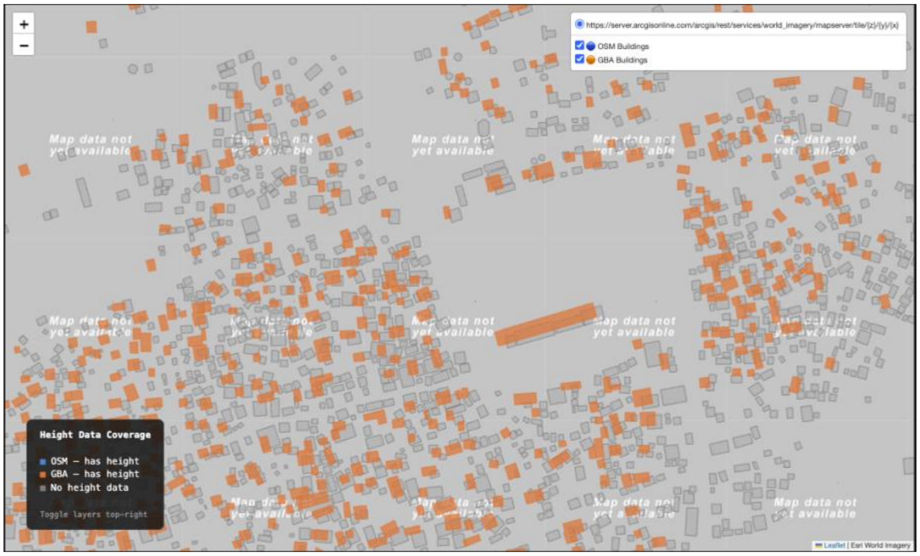

Image 1: Comparison of OSM vs GBA footprints on Tawila

5.1.4 Interpretation

Tawila has benefited from a targeted OSM mapping effort by community mapathons. These are known to have taken place in Sudan in response to the displacement crisis. This accounts for OSM's comparatively higher building count and finer-grained footprint delineation in this area. This is also visually supported by the map, where several building clusters that OSM represent multiple distinct, individually traced polygons appear in GBA as a smaller number of larger, merged polygons. This is also consistent with GBA’s known limitation, documented in the GBA paper itself, that its PSR-derived footprints (at 3m satellite resolution) can merge adjacent structures into a single polygon rather than resolving them individually.

This suggests that for building footprint polygon precision specifically, human-mapped OSM data may outperform GBA in areas that have received focused community mapping attention. However, GBA still offers a clear advantage in height data, which OSM does not provide.

5.2 Area of Interest: Sobat Corridor, South Sudan

5.2.1 Site Context

The Sobat corridor is a large, sparsely populated rural area in South Sudan. To reduce processing time, a smaller AOI was selected for the corridor, Malakal, the regional capital of Upper Nile State and a major humanitarian hub, anchoring the corridor's southern/western end near the White Nile confluence (the corridor itself extends northeast from Malakal toward the Ethiopian border). The area is a humanitarian hub for hosting displaced people.

Coordinates: 9.52, 31.66

5.3.2 Data Comparison

Summary of OSM, GBA and Google 2.5D building footprint datasets statistics for Malakal:

Metric	OSM	GBA	Google 2.5D
Building count	16,241	24,328	N/A
Total footprint area (km²)	0.9911	1.2261	0.9963
Buildings with height data	0%	100%	71.2%

Note: Google 2.5D has no discrete building count, since it is a raster rather than a vector dataset. An earlier version of this table listed 59,927 in this cell, which was in fact the summed building-presence value (a probability-weighted pixel total, not a count of buildings) and has been corrected to N/A here.

5.2.3 Visual Comparison

Image 2: Maps of GBA vs OSM footprints on Malakal with and without basemap respectively

5.2.4 Interpretation

The Sobat Corridor presents a distinct data profile relative to the other two AOIs. Where Budapest represents a best-case, training-data-rich scenario and Tawila shows OSM outpacing GBA at the individual-building level, the Sobat Corridor represents the opposite extreme: a sparsely populated, low-connectivity rural region where OSM coverage remains limited. Here, GBA identifies substantially more buildings than OSM (+49%), suggesting its satellite-derived pipeline is doing genuine completeness work rather than merely replicating or refining existing OSM data, as appeared to be the case in Budapest.

Google 2.5D's built-up area estimate for this AOI (0.9963 km²) sits remarkably close to OSM's (0.9911 km²), a stronger agreement than observed between either dataset or GBA (1.2261 km²) here. This is a useful cross-validation signal for this specific AOI, though given the structural differences between how each dataset derives area (traced polygons for OSM/GBA versus summed pixel-level confidence for Google 2.5D), it should be read as a point of agreement worth noting rather than proof that either figure is more accurate.

5.3 Area of Interest: Budapest, Hungary Europe

5.3.1 Site Context

Budapest is a city located in Hungary. This AOI was selected as a comparison use case from which the model had maximum training data as reported to perform well.

Coordinates: 47.50, 19.05

5.3.2 Data Comparison

Summary of OSM vs. GBA building counts and footprint areas for Budapest.

Note that there is no comparison with Google 2.5D because the dataset coverage does not include Europe.

Metric	OSM	GBA	Difference
Building count	39,881	45,952	+15%
Total footprint area (km²)	39.93	41.24	+3.3%
Buildings with height data	1.6%	100%	+98%

5.3.3 Visual Comparison

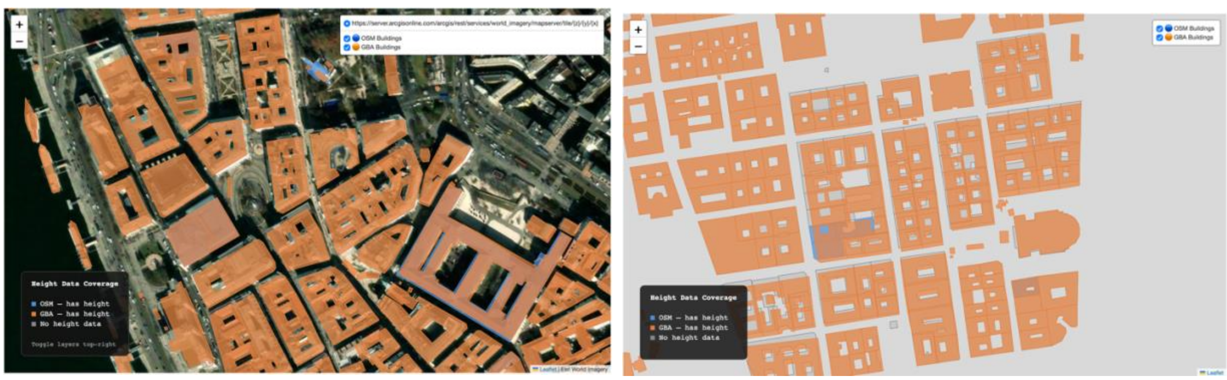

Image 3: Maps of GBA vs OSM footprints on Budapest city, with and without basemap respectively

5.3.4 Interpretation

Budapest was selected as a benchmark AOI representing a near-ideal operating condition for GBA, given that its training data for building height and footprint generation drew heavily on European sources. The results support this framing. Visually, GBA's footprints closely track individual building outlines against satellite imagery, including complex, non-rectangular block structures, in clear contrast to the generalized, merged polygons observed in less-mapped AOIs such as Tawila. A likely explanation is that OSM served as the primary base layer in GBA's polygon fusion process for this region, as described by the paper, OSM as the default base layer for all continents except South America and Africa. This meaning GBA's footprints in Budapest are plausibly derived substantially from OSM itself rather than GBA's independent satellite-derived pipeline. This is consistent with the close spatial overlap observed between OSM and GBA polygons in the AOI.

Despite this strong footprint agreement, height data completeness diverges sharply between the two sources: OSM provides height values for only 1.6% of buildings, while GBA reaches 100% completeness. This confirms a pattern consistent across AOIs assessed in this study that OSM's strength lies in community-mapped footprint geometry, while GBA's core operational advantage remains height data, which OSM does not meaningfully provide even in well-mapped urban contexts. Building count and total footprint area were broadly comparable between the two datasets (39,881 vs. 45,952 buildings; 39.93 km² vs. 41.24 km² respectively), reflecting the relatively strong alignment expected in a region well-represented in GBA's training and fusion inputs.

This AOI should be treated as a favorable benchmark rather than a representative case: it illustrates GBA's performance ceiling in a well-mapped, training-data-rich European context, and is not necessarily indicative of expected performance in regions with sparse OSM coverage or absent from GBA's validation set, such as Sub-Saharan Africa.

6. Known Limitations

6.1 Training Data Gaps

- No LiDAR validation data available for Africa — height model neither trained nor validated there
- Minimal training representation in Asia (2 cities) and South America (1 city) vs. Europe (109) and North America (39)

6.2 Height Accuracy

GBA height Accuracy differs among regions, as shown on the table below. Note that GBA's error is reported as RMSE (Root Mean Square Error), while Google 2.5D's error is reported as MAE (Mean Absolute Error) these are not directly comparable metrics. Additionally, Google 2.5D's 1.5m MAE is the reported global value applied to all regions but validated in North America, Europe and Japan.

Region	GBA Height RMSE (m)	Google 2.5D Height Error _MAE(m)
Africa	N/A	1.5
Oceania	1.5	1.5
Europe	4.1	1.5
North America	5.3	1.5
Asia	5.9	1.5
South America	8.9	1.5

6.3 Data Currency

The dataset reflects a ~2019-era snapshot (with fused sources spanning 2014–2021), not real-time conditions. Not suitable as a standalone source for monitoring recent/ongoing construction or displacement without supplementary data.

6.4 Height Underestimation in Dense/High-Rise Areas

The height estimation model tends to underpredict in dense, vertical urban environments, observed in South American and Asian test cities. Less relevant for low-rise/informal settlement contexts, but worth noting if applicable to your AOIs.

6.5. Confidence threshold sensitivity (Google 2.5D)

Google 2.5D's area figure is more sensitive to an analyst's arbitrary methodological choice than OSM or GBA's polygon-based counts are, since the latter have no equivalent threshold parameter. A large spread across thresholds should be reported as a source of uncertainty alongside the headline built-up area number, not silently resolved by picking whichever threshold looks best

7. Guidance for Replication

The workflow used in this report is documented in the accompanying notebook (see Appendix A) and can be adapted to additional areas of interest.

8. Conclusion

This report assessed the operational value of TU Munich's GlobalBuildingAtlas and Google's Open Buildings 2.5D Temporal dataset, alongside OpenStreetMap, for building-level population estimation in support of MSF's humanitarian operations. Across Tawila, Budapest, and the Sobat Corridor, as expected, one finding held without exception: GBA and Google 2.5D both provide substantial building height data where OSM provides essentially none. This remains the clearest, most consistently demonstrated operational advantage either dataset offers, and it is the foundation for any volume-based population proxy this report has discussed. Interpretation should, however, be supported by visual comparison and context.

Two methodological contributions of this report are worth restating. First, the empirical test of GBA's tile-boundary behavior, not documented in the source paper, found that buildings crossing a tile edge are duplicated safely, with identical IDs and geometry, resolvable through a single deduplication step. Second, the threshold sensitivity check applied to Google 2.5D showed that its built-up area estimates can vary considerably depending on the confidence threshold chosen, a source of uncertainty that has no equivalent in OSM or GBA's polygon-based counts and should be reported alongside any built-up area figure derived from it.

Taken together, these findings support a cautious, context-dependent approach to adopting dataset into MSF's existing population-estimation workflows. Both GBA and Google 2.5D add real value as height-data sources; neither has been validated with ground-truth data in Africa, the region most central to MSF's operations; and GBA's 2019-era imagery means neither dataset, on its own, can support monitoring of recent or ongoing displacement. Before being used to inform operational decisions, figures derived from either dataset should be cross-checked against locally available reference data i.e. census figures, field registration counts, or WorldPop estimates, rather than applied as standalone outputs. Extending this comparison to the remaining areas of interest named in the original scope and validating a volume-based population estimate against a known reference population, remain the most valuable next steps for continuing this work. Additionally, running the GBA model on recent PlaneScope images, given the technical and computational resources, will ensure its application to recent or ongoing operations.

9. Appendices

Appendix A: Notebook / Code Reference

GitHub link to Notebook

Appendix B: Data Access and Licensing Notes

- GBA.ODbLPolygon (OSM/Microsoft-derived): Open Database License (ODbL)
- GBA.LoD1 (Part II — Google Open Buildings/CLSM-derived): CC BY-NC 4.0
- GBA processing code: MIT License with Commons Clause (non-commercial use only)
- Google Open Buildings 2.5D Temporal: Creative Commons Attribution 4.0 (CC BY 4.0)

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